Global Economics

Grade 10 – Term 1 – C1–C3 Classroom Activity – Version 2

A decision under uncertainty selects an alternative A_i before a state S_j is known. Each pair (A_i, S_j) determines a payoff

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j),$$

placed in row A_i, column S_j. The objective is to construct the complete payoff table.

Revenue and cost are

$$R = q(A_i, S_j)\big[r(A_i) + r(S_j)\big] + f_r(A_i) + f_r(S_j),$$

$$C = q(A_i, S_j)\big[c(A_i) + c(S_j)\big] + f_c(A_i) + f_c(S_j),$$

$$P(A_i, S_j) = q(A_i, S_j)\big[r(A_i) + r(S_j) - c(A_i) - c(S_j)\big] + f_r(A_i) + f_r(S_j) - f_c(A_i) - f_c(S_j).$$

Variable components are per-unit rates multiplied by $q(A_i, S_j)$; fixed components are not. A_i, S_j, and (A_i, S_j) indicate dependence on the alternative, state, or both. Revenue rates are gross revenue; costs are calculated separately. Missing components are zero; directly given R or C require no decomposition.

Context Descriptions

For each problem, **(C1)** *identify the decision alternatives, the states of nature, the uncertain event, and the consequence.* Then **(C3)** *calculate revenue, cost, and profit for every alternative–state pair and construct the payoff table.*

1. **Direct case.** A planetarium must select an immersive show or an astronomy workshop for its holiday programme before evening cloud cover is known. Evenings may be clear or cloudy, affecting interest in indoor astronomy activities. All amounts are total revenue and total cost, in thousand USD. For the immersive show, revenue and cost are 64 and 18 in clear evenings, and 35 and 22 in cloudy evenings. For the astronomy workshop, revenue and cost are 53 and 24 in clear evenings, and 46 and 23 in cloudy evenings. These totals include all revenue and costs for the programme.
2. **One-component problem.** An equipment rental firm must choose to offer lighting rigs or sound rigs before the external festival calendar is confirmed. The calendar may be full or reduced by organiser cancellations. Quantities are measured in rig-weeks: one rig rented for one week. Under a full and a reduced calendar, respectively, lighting rentals total 7 and 3 rig-weeks, and sound rentals total 6 and 2 rig-weeks. Gross rental revenue is 14 thousand USD per rig-week for lighting rigs and 18 for sound rigs. Total operating cost per rig-week, covering equipment hire, setup and staffing, is 6 thousand USD with a full calendar and 9 with a reduced calendar, for either rig type. There are no fixed amounts or other components.

Context Descriptions (continued)

3. **Two-component problem.** A language school must choose a group course or a small-group course before overseas arrivals are known. Arrivals may be many or few. Enrolments, in hundreds of students, are 8 and 5 for the group course and 7 and 4 for the small-group course under many and few arrivals, respectively. Gross tuition rates are 9 and 12 thousand USD per hundred students for the two formats. An external education fund adds gross teaching payments of 4 thousand USD per hundred students with many arrivals and 2 with few arrivals, for either format. Teaching and materials cost 5 and 7 thousand USD per hundred students for the two formats. Fixed room-booking costs, paid once for the whole course, are 8 thousand USD with many arrivals and 14 with few arrivals, for either format. All other components are zero.
4. **Three-component problem.** An indoor climbing centre must choose bouldering sessions or rope-climbing sessions before holiday weather is known. The period may be wet or dry. Quantities, in hundreds of participant-visits, are 9 and 4 for bouldering and 11 and 7 for rope climbing under wet and dry weather, respectively. Gross session fees are 10 and 13 thousand USD per hundred participant-visits for the two types. A tourism partner adds gross referral payments of 4 and 1 thousand USD per hundred participant-visits in wet and dry weather, for either type. Fixed sponsorship revenue, received once, is 18 thousand USD for bouldering and 27 for rope climbing. Instruction and equipment costs are 6 and 7 thousand USD per hundred participant-visits for the two types. Weather-related cleaning and energy add costs of 2 and 4 thousand USD per hundred participant-visits in wet and dry weather. Fixed building-operation costs are 12 and 17 thousand USD in wet and dry weather, paid once. Both weather-based costs apply to either type; all other components are zero.

1 Solutions: Revenue, Costs, Profit, and Payoff Tables

Problems:

1.1. Planetarium programme (C1–C3)

1.2. Festival equipment rental (C1–C3)

1.3. Language course format (C1–C3)

1.4. Climbing centre sessions (C1–C3)

1.1 Exercise 1: Planetarium programme (direct case)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Immersive show	A_2: Astronomy workshop
States of nature	S_1: Clear evenings	S_2: Cloudy evenings
Uncertain event	Evening cloud cover revealed after the programme choice	
Consequences	The selected activity under the realised external conditions produces a profit or loss, measured in thousand USD for each alternative–state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Immersive show; A_2: Astronomy workshop	Alternative labels, selected before the state is known; row order.
S_j	S_1: Clear evenings; S_2: Cloudy evenings	External states, revealed after the choice; column order.
$R(A_i, S_j)$	$(1,1)$: 64 thousand USD; $(1,2)$: 35 thousand USD; $(2,1)$: 53 thousand USD; $(2,2)$: 46 thousand USD	Supplied total revenue; depends on both alternative and state. No decomposition.
$C(A_i, S_j)$	$(1,1)$: 18 thousand USD; $(1,2)$: 22 thousand USD; $(2,1)$: 24 thousand USD; $(2,2)$: 23 thousand USD	Supplied total cost; depends on both alternative and state. No decomposition.

The total revenue and total cost are supplied directly for every cell, so

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1) - C(A_1,S_1) = P(A_1,S_1)$	$R(A_1,S_2) - C(A_1,S_2) = P(A_1,S_2)$
A_2	$R(A_2,S_1) - C(A_2,S_1) = P(A_2,S_1)$	$R(A_2,S_2) - C(A_2,S_2) = P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$64 - 18 = 46$	$35 - 22 = 13$
A_2	$53 - 24 = 29$	$46 - 23 = 23$

Final payoff table (profit in thousand USD).

Alternative	Clear evenings (S_1)	Cloudy evenings (S_2)
Immersive show (A_1)	46	13
Astronomy workshop (A_2)	29	23

1.2 Exercise 2: Festival equipment rental (one component)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Lighting rigs	A_2: Sound rigs
States of nature	S_1: Full festival calendar	S_2: Reduced festival calendar
Uncertain event	The external festival calendar revealed after the equipment choice	
Consequences	The selected activity under the realised external conditions produces a profit or loss, measured in thousand USD for each alternative–state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Lighting rigs; A_2: Sound rigs	Alternative labels, selected before the state is known; row order.
S_j	S_1: Full festival calendar; S_2: Reduced festival calendar	External states, revealed after the choice; column order.
$q(A_i, S_j)$	$(1,1)$: 7 rig-weeks; $(1,2)$: 3 rig-weeks; $(2,1)$: 6 rig-weeks; $(2,2)$: 2 rig-weeks	Quantity depends on both alternative and state.
$r(A_i)$	14; 18 thousand USD/rig-week	Gross rental revenue rate for lighting; sound rigs. Alternative only. Values follow index order.
$c(S_j)$	6; 9 thousand USD/rig-week	Total operating cost rate with a full; reduced calendar. State only; applies to both alternatives. Values follow index order.

All amounts below are in thousand USD. Fixed amounts are applied once; all unlisted components are zero. Revenue has one alternative-based variable component, while cost has one state-based variable component:

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times r(A_1)$ $= R(A_1, S_1)$	$q(A_1, S_2) \times r(A_1)$ $= R(A_1, S_2)$
A_2	$q(A_2, S_1) \times r(A_2)$ $= R(A_2, S_1)$	$q(A_2, S_2) \times r(A_2)$ $= R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	7×14 $= 98$	3×14 $= 42$
A_2	6×18 $= 108$	2×18 $= 36$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times c(S_1)$ $= C(A_1, S_1)$	$q(A_1, S_2) \times c(S_2)$ $= C(A_1, S_2)$
A_2	$q(A_2, S_1) \times c(S_1)$ $= C(A_2, S_1)$	$q(A_2, S_2) \times c(S_2)$ $= C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	7×6 $= 42$	3×9 $= 27$
A_2	6×6 $= 36$	2×9 $= 18$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$98 - 42$ $= 56$	$42 - 27$ $= 15$
A_2	$108 - 36$ $= 72$	$36 - 18$ $= 18$

Final payoff table (profit in thousand USD).

Alternative	Full festival calendar (S_1)	Reduced festival calendar (S_2)
Lighting rigs (A_1)	56	15
Sound rigs (A_2)	72	18

1.3 Exercise 3: Language course format (two components)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Group course	A_2: Small-group course
States of nature	S_1: Many overseas arrivals	S_2: Few overseas arrivals
Uncertain event	Overseas arrivals revealed after the course-format choice	
Consequences	The selected activity under the realised external conditions produces a profit or loss, measured in thousand USD for each alternative–state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Group course; A_2: Small-group course	Alternative labels, selected before the state is known; row order.
S_j	S_1: Many overseas arrivals; S_2: Few overseas arrivals	External states, revealed after the choice; column order.
$q(A_i, S_j)$	$(1,1)$: 8 hundreds of students; $(1,2)$: 5 hundreds of students; $(2,1)$: 7 hundreds of students; $(2,2)$: 4 hundreds of students	Quantity depends on both alternative and state.
$r(A_i)$	9; 12 thousand USD/100 students	Gross tuition rate for group; small-group courses. Alternative only. Values follow index order.
$r(S_j)$	4; 2 thousand USD/100 students	Gross education-fund teaching payment rate with many; few arrivals. State only; applies to both formats. Values follow index order.
$c(A_i)$	5; 7 thousand USD/100 students	Teaching and materials cost rate for group; small-group courses. Alternative only. Values follow index order.
$f_c(S_j)$	8; 14 thousand USD	Fixed room-booking total with many; few arrivals. State only; paid once for either format. Values follow index order.

All amounts below are in thousand USD. Use the numerical count of hundreds as q with rates per hundred; do not multiply by 100 again. Fixed amounts are applied once; all unlisted components are zero. Revenue combines alternative-based and state-based variable components. Cost combines an alternative-based variable component and a state-based fixed component:

$$R = q(A_i, S_j)[r(A_i) + r(S_j)], \quad C = q(A_i, S_j)c(A_i) + f_c(S_j), \quad P = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times [r(A_1)+r(S_1)] = R(A_1,S_1)$	$q(A_1,S_2)\times [r(A_1)+r(S_2)] = R(A_1,S_2)$
A_2	$q(A_2,S_1)\times [r(A_2)+r(S_1)] = R(A_2,S_1)$	$q(A_2,S_2)\times [r(A_2)+r(S_2)] = R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times [9+4] = 104$	$5\times [9+2] = 55$
A_2	$7\times [12+4] = 112$	$4\times [12+2] = 56$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(A_1) + f_c(S_1) = C(A_1,S_1)$	$q(A_1,S_2)\times c(A_1) + f_c(S_2) = C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(A_2) + f_c(S_1) = C(A_2,S_1)$	$q(A_2,S_2)\times c(A_2) + f_c(S_2) = C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times 5 + 8 = 48$	$5\times 5 + 14 = 39$
A_2	$7\times 7 + 8 = 57$	$4\times 7 + 14 = 42$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$104-48$ $=56$	$55-39$ $=16$
A_2	$112-57$ $=55$	$56-42$ $=14$

Final payoff table (profit in thousand USD).

Alternative	Many overseas arrivals (S_1)	Few overseas arrivals (S_2)
Group course (A_1)	56	16
Small-group course (A_2)	55	14

1.4 Exercise 4: Climbing centre sessions (three components)

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Bouldering sessions	A_2: Rope-climbing sessions
States of nature	S_1: Wet holiday period	S_2: Dry holiday period
Uncertain event	Holiday weather revealed after the session-type choice	
Consequences	The selected activity under the realised external conditions produces a profit or loss, measured in thousand USD for each alternative–state pair.	



C3 - Calculating profit and building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Bouldering sessions; A_2: Rope-climbing sessions	Alternative labels, selected before the state is known; row order.
S_j	S_1: Wet holiday period; S_2: Dry holiday period	External states, revealed after the choice; column order.
$q(A_i, S_j)$	$(1,1)$: 9 hundreds of participant-visits; $(1,2)$: 4 hundreds of participant-visits; $(2,1)$: 11 hundreds of participant-visits; $(2,2)$: 7 hundreds of participant-visits	Quantity depends on both alternative and state.
$r(A_i)$	10; 13 thousand USD/100 participant-visits	Gross session fee rate for bouldering; rope climbing. Alternative only. Values follow index order.
$r(S_j)$	4; 1 thousand USD/100 participant-visits	Gross tourism referral rate in wet; dry weather. State only; applies to both types. Values follow index order.
$f_r(A_i)$	18; 27 thousand USD	Fixed sponsorship total for bouldering; rope climbing. Alternative only; received once. Values follow index order.
$c(A_i)$	6; 7 thousand USD/100 participant-visits	Instruction and equipment cost rate for bouldering; rope climbing. Alternative only. Values follow index order.
$c(S_j)$	2; 4 thousand USD/100 participant-visits	Additional cleaning and energy cost rate in wet; dry weather. State only; applies to both types. Values follow index order.
$f_c(S_j)$	12; 17 thousand USD	Fixed building-operation total in wet; dry weather. State only; paid once for either type. Values follow index order.

All amounts below are in thousand USD. Use the numerical count of hundreds as q with rates per hundred; do not multiply by 100 again. Fixed amounts are applied once; all unlisted components are zero. Revenue has alternative- and state-based variable components and an alternative-based fixed component. Cost has alternative- and state-based variable components and a state-based fixed component:

$$R = q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i),$$
$$C = q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(S_j),$$
$$P = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$9\times$ $[10+4]$ $+18$ $=144$	$4\times$ $[10+1]$ $+18$ $=62$
A_2	$11\times$ $[13+4]$ $+27$ $=214$	$7\times$ $[13+1]$ $+27$ $=125$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$9\times$ $[6+2]$ $+12$ $=84$	$4\times$ $[6+4]$ $+17$ $=57$
A_2	$11\times$ $[7+2]$ $+12$ $=111$	$7\times$ $[7+4]$ $+17$ $=94$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$144-84$ $=60$	$62-57$ $=5$
A_2	$214-111$ $=103$	$125-94$ $=31$

Final payoff table (profit in thousand USD).

Alternative	Wet holiday period (S_1)	Dry holiday period (S_2)
Bouldering sessions (A_1)	60	5
Rope-climbing sessions (A_2)	103	31